

Evaluation

Introduction to Information Retrieval

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Retrieval Models: Keyword Matching

$$\text{Score}(d,Q) = \begin{cases} \sum_{t \in Q} f(d, t) \\ \prod_{t \in Q} g(d, t) \\ \dots \end{cases}$$

Query: *Vehicle Smoke Pollution*

$$\text{Score}(d,Q) = f(d, \text{Vehicle}) + f(d, \text{Smoke}) + f(d, \text{Pollution})$$

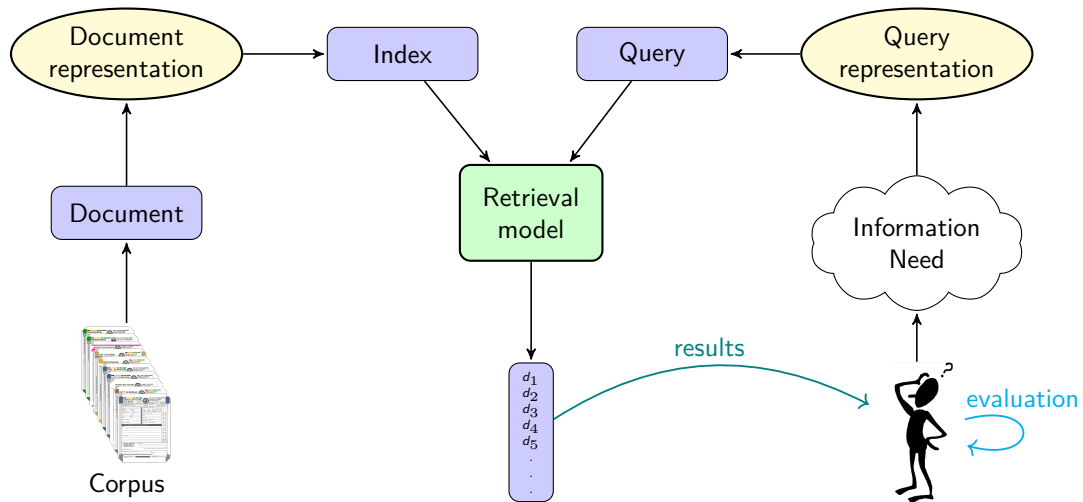
- $f(\cdot)$, $g(\cdot)$ - term weighting functions.

Retrieval Models: Term weights

$$w(q, d)_{LM} = \frac{tf(q, d) + \mu p(q|C)}{|d| + \mu} \quad (1)$$

$$w(q, d)_{BM25} = \frac{D - df(q) + 0.5}{df(q) + 0.5} \frac{tf(q, d)(k_1 + 1)}{tf(q, d) + k_1(1 - b + b \frac{|d|}{avgdl})} \quad (2)$$

Information Retrieval - A graphical representation



Comparing Retrieval Models: Motivation

- Which is better: Heap sort or Bubble sort?

Comparing Retrieval Models: Motivation

Model 1



BM25

Model 2



LM

Model 3



DFR

Comparing Retrieval Models: Motivation

Model 1



BM25

Model 2



LM

Model 3



DFR

Criteria for comparison

- Time.
- Quality of Ranked list.

Motivation

- IR is an *empirical* discipline.
- Intuition can be wrong!
 - ▶ “sophisticated” techniques need not be the best:
e.g. rule-based stemming vs. statistical stemming

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- Intuition can be wrong!
 - ▶ “sophisticated” techniques need not be the best:
e.g. rule-based stemming vs. statistical stemming
- Proposed techniques need to be validated and compared to existing techniques.

Cranfield Method (CLEVERDON ET AL., 60s)

Benchmark data

- Document collection .
- Query / topic collection .
- Relevance judgments - information about which document is relevant to which query.

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syllabus

question paper

correct answers

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Assumptions

- relevance of a document to a query can objectively be determined;
- all relevant documents in the collection are known;
- all relevant documents contribute equally to the performance measures;
- relevance of a document is independent of the relevance of other documents.

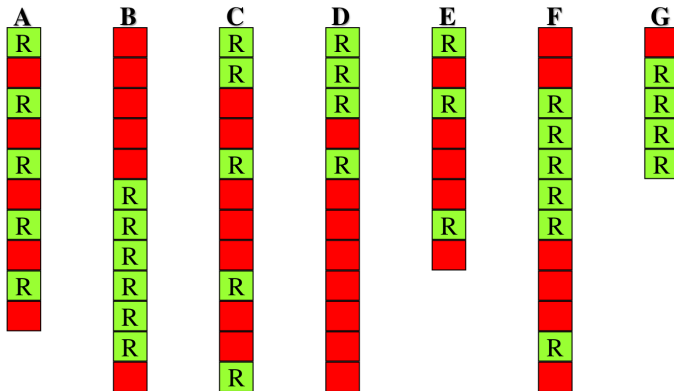
Evaluation Metrics

Background

- User has an information need.
- Information need is converted into a **query**.
- Documents are **relevant** or **non-relevant**.
- Ideal system retrieves all and only the relevant documents.

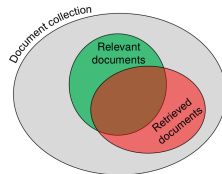
Compare a Set of Engines

- Query $Q = \{q_1, q_2, \dots\}$
- There are 8 relevant documents in the collection.



Set-based Evaluation Metrics

	Relevant	Non-relevant	Total
Retrieved	True Positives (TP)	False Positives (FP)	Retrieved
Not Retrieved	False Negatives (FN)	True Negatives (TN)	Not Retrieved
Total	Relevant	Non-relevant	All Documents



Perfect System

- TP = All relevant docs
- FP = 0
- FN = 0
- TN = All non-relevant docs

Useless System

- Returns everything: FP = All non-relevant
- Returns nothing: FN = All relevant

$$\begin{aligned}\text{Recall} &= \frac{\#(\text{relevant retrieved})}{\#(\text{relevant})} \\ &= \frac{\#TP}{\#TP + \#FN}\end{aligned}$$

$$\begin{aligned}\text{Precision} &= \frac{\#(\text{relevant retrieved})}{\#(\text{retrieved})} \\ &= \frac{\#TP}{\#TP + \#FP}\end{aligned}$$

Combining P and R

- Harmonic mean of precision and recall:

$$\mathbf{F1} = \frac{2 \cdot P \cdot R}{P + R}$$

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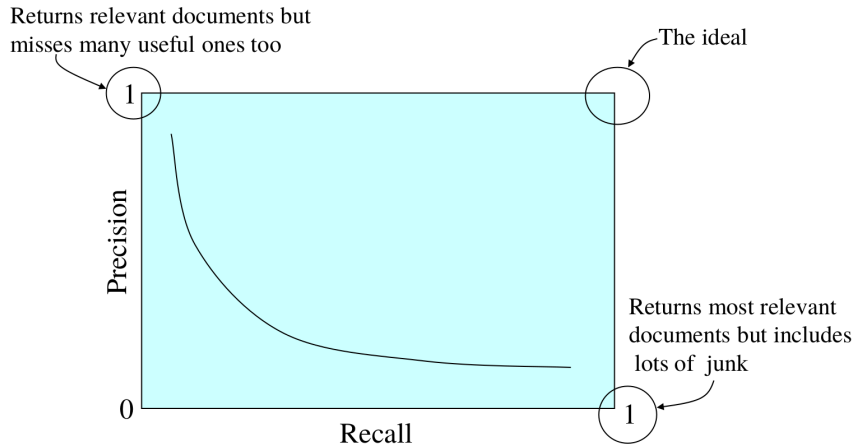
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- ▶ β controls relative importance of P and R
 - ★ $\beta = 1 \rightarrow$ precision and recall equally important (F1);
 - ★ $\beta = 3 \rightarrow$ recall three times more important than precision.

Trade-off between P and R

- **Precision:** measures the ability to retrieve relevant docs at top-ranks.
- **Recall:** measures the ability to retrieve all relevant items in the corpus.
- If we retrieve more documents:
 - ▶ higher chance to find all relevant docs $\rightarrow R \uparrow$
 - ▶ higher chance to find more irrelevant docs $\rightarrow P \downarrow$

Trade-off between P and R

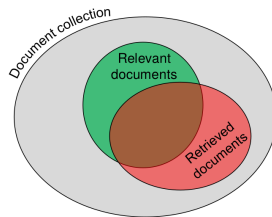


slide courtesy - W. Magdy (Univ. of Edinburgh)

Accuracy

- Accuracy $\rightarrow$ fraction of docs classified correctly.

$$\begin{aligned}\text{Accuracy} &= \frac{\# \text{correct-retrieved} + \# \text{correct not-retrieved}}{\# \text{total docs.}} \\ &= \frac{\# \text{TP} + \# \text{TN}}{\# \text{TP} + \# \text{FP} + \# \text{TN} + \# \text{FN}}\end{aligned}$$



Why Accuracy Fails in IR

The Class Imbalance Problem

Typical Collection Statistics

- Collection size: 1,000,000 documents
- Relevant documents for a query: 50 documents
- Non-relevant documents: 999,950 documents

Useless System A

Returns **NOTHING**

$$\text{Accuracy} = \frac{0+999,950}{1,000,000} = 99.995\%$$

Useless System B

Returns **EVERYTHING**

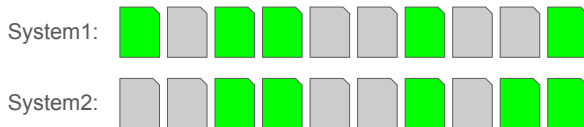
$$\text{Accuracy} = \frac{50+0}{1,000,000} = 0.005\%$$

Key Insight

- Accuracy is meaningless when classes are imbalanced.
- We need metrics sensitive to the rare class (relevant documents).

Incorporating Ranking in Precision/Recall

- Let's assume, for a given query, there are 5 **Relevant** documents.



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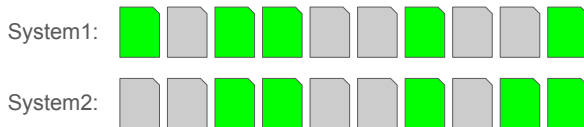
System1:



- Both retrieved 10 docs, only 5 are **relevant**.
- P, R and F are the **same** for both systems.
- Should their performances considered **equal**?

Incorporating Ranking in Precision/Recall

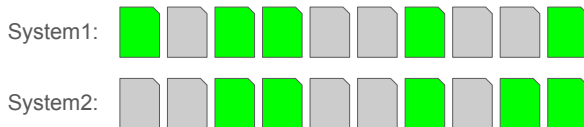
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- We can compute precision till some rank position.

Precision@K

- **K** $\rightarrow$ a measuring parameter $\rightarrow$ measure precision upto rank **K**.
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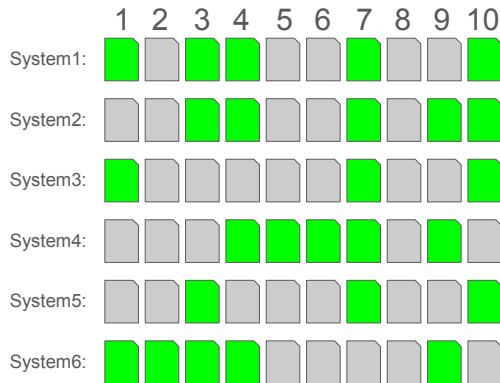
Precision@K

- $K \rightarrow$ a measuring parameter $\rightarrow$ measure precision upto rank K .
 - Applicable for web searches: most people only check top K results.
 - $K \rightarrow$ fixed for all queries.
- Two systems should always be compared based on the *same set of queries*.

Compare a Set of Engines

using P@5

- Query $Q = \{q_1, q_2, \dots\}$
- There are 8 relevant documents in the collection.



R-precision

- For a query with known **R** relevant documents:
 - ▶ R-precision $\rightarrow$ precision at rank **R** ($P@R$)
- **R** $\rightarrow$ different for each query.

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 - ▶ R-precision $\rightarrow$ precision at rank **R** ($P@R$)
- **R** $\rightarrow$ different for each query.
- A measurement of *idealness* of the system $\rightarrow$ getting all relevant documents in the top ranks.

R-Precision in Action

A Concrete Example

Scenario

- Query has **R = 8** known relevant documents
- System retrieves documents in some order
- We calculate precision at rank 8

System A

Ranking: R, N, R, R, N, N, R, N, R, R, R, R

At rank 8: 4 relevant docs found

R-Precision = $4/8 = 0.5$

System B

Ranking: R, R, R, R, R, R, R, R, N, N, N, N

At rank 8: 8 relevant docs found

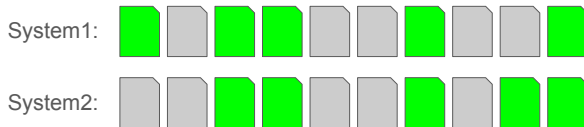
R-Precision = $8/8 = 1.0$

Interpretation

System B is **ideal** - it found all relevant documents in the top R positions. System A missed half of them.

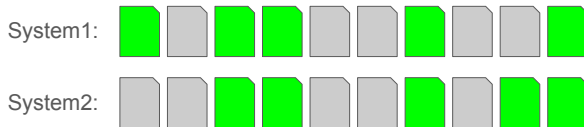
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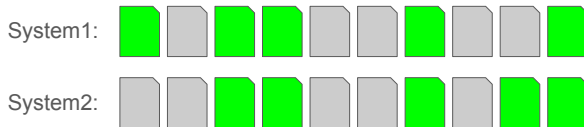
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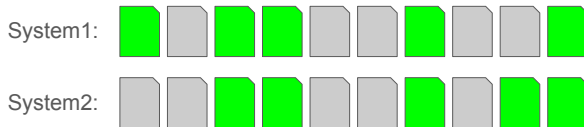
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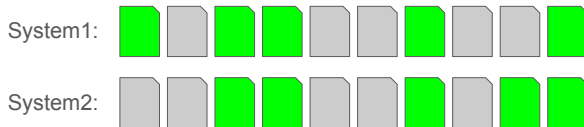
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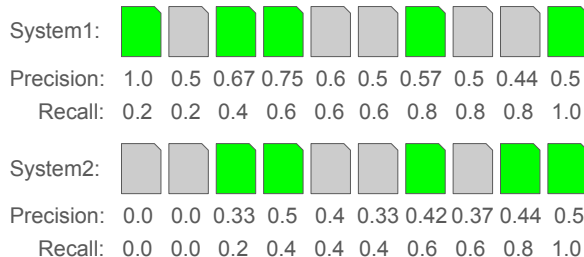
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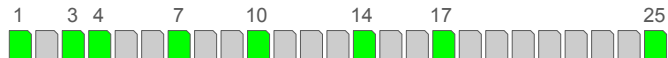
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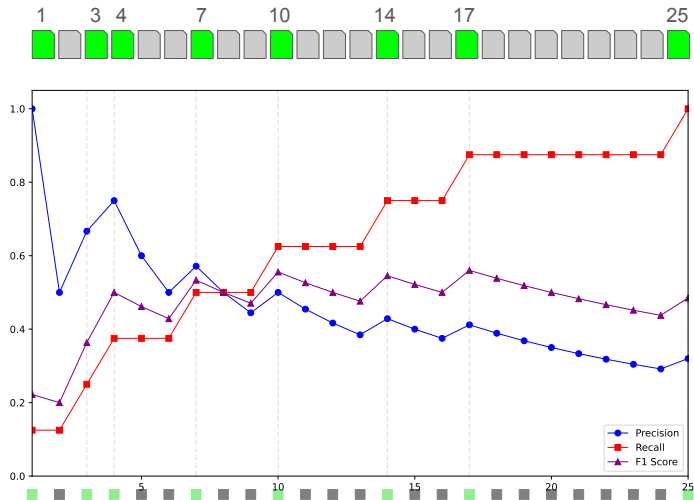
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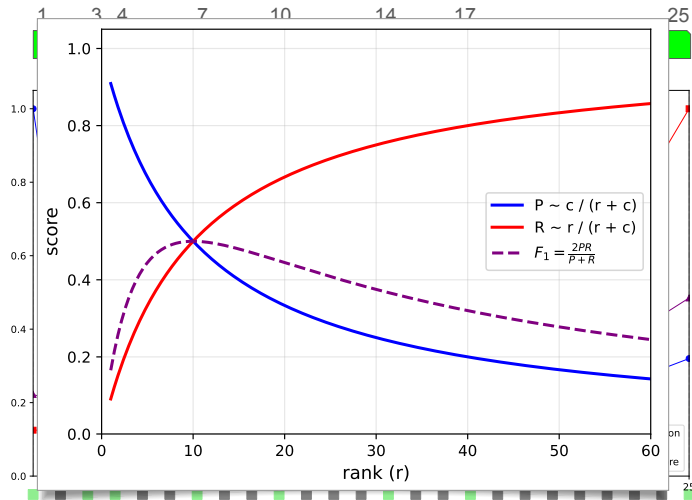
Precision/Recall/F1 Vs. Rank



Precision/Recall/F1 Vs. Rank



Precision/Recall/F1 Vs. Rank



(non-interpolated) Average Precision

- Precision at each point a new relevant document is retrieved. $\rightarrow$ Average.

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- **Precision** at each point a new relevant document is retrieved. → **Average**.



Rank	Type	Recall	Precision
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(5 relevant docs. in all)

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- A mix between precision and recall.
- Highly focus on finding relevant document as early as possible.
- Most commonly used evaluation metric for most IR search tasks.
- Uses binary relevance: rel = 0/1.

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Total 5 relevant documents.



2 relevant documents not retrieved.

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- AP - Computed per-query.

From AP (single query) to MAP (set of queries)

Mean AP

Mean Average Precision (MAP) is simply the **mean** of AP scores over all queries in query set $\mathcal{Q}$.

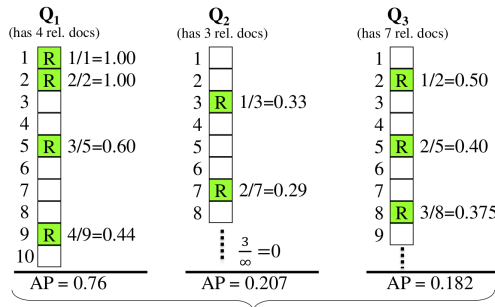
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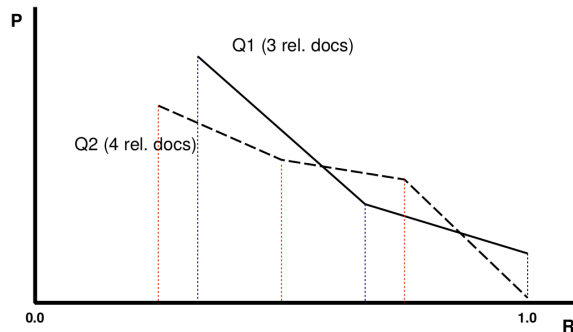
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$$\text{MAP} = (0.76 + 0.207 + 0.182) / 3 = \mathbf{0.383}$$

Average Precision

how to combine across set of queries

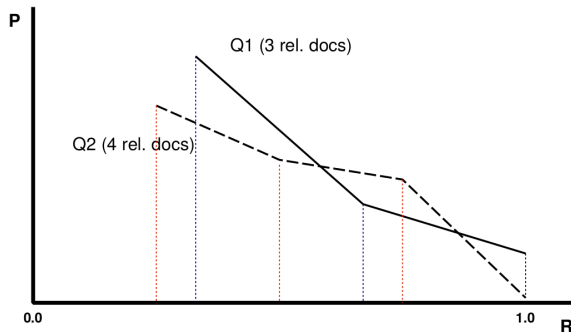


$$AvgP = \frac{1}{N_{Rel}} \sum_{d_i \in Rel} \frac{i}{Rank(d_i)}$$

- Plot Recall-Precision curve per-query.
- Recall points correspond to $\frac{1}{N_{Rel}}$.
- N_{Rel} different for different queries.

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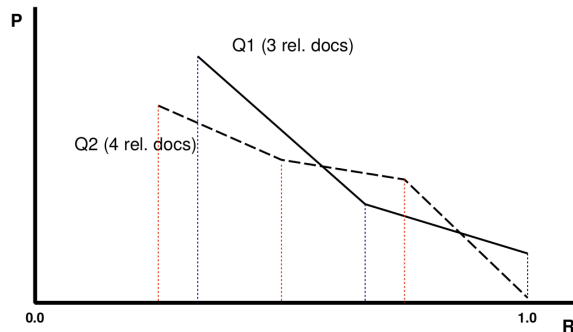


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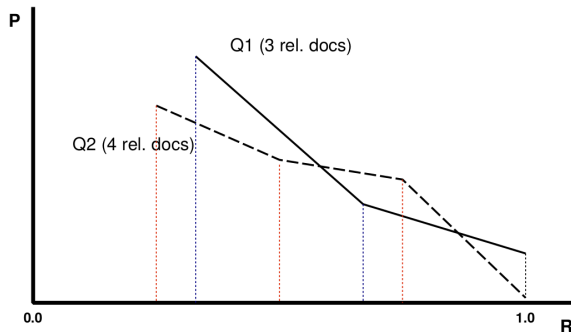


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- Average *interpolated precision* at standard recall levels.

11-pt interpolated average precision

$$P_{int}(r) = \max_{r' \geq r} P(r')$$

at each 11 standard recall level $r_i \in \{0.0, 0.1, \dots, 1.0\}$



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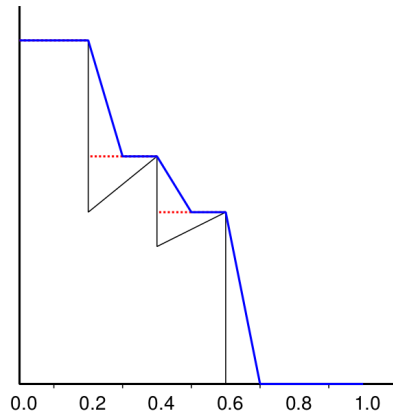
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Geometric MAP

GMAP

System 1		
Rank	Topic 1	Topic 2
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4	Non-relevant	Non-relevant
5	Non-relevant	Non-relevant
AP	0.5	0.5

Geometric MAP

GMAP

System 1		
Rank	Topic 1	Topic 2
1	Non-relevant	Non-relevant
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MAP	0.5	

System 2		
Rank	Topic 1	Topic 2
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2	Non-relevant	Non-relevant
3	Non-relevant	Non-relevant
4	Non-relevant	Non-relevant
5	Non-relevant	Non-relevant

Geometric MAP

GMAP

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MAP	0.5	

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MAP	0.5	

Geometric MAP

$$GMAP(AP_1, \dots, AP_n) = \left(\prod_{i=1}^n (AP_i + \epsilon) \right)^{1/n} - \epsilon$$

- ϵ - a constant to eliminate pathologies due to one of the AP_i s being 0.

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4	Non-relevant	Non-relevant
5	Non-relevant	Non-relevant
AP	1.0	0.0
GMAP	$\sqrt{1.01 * 0.01} \approx 0.10$	

Mean Reciprocal Rank

- Evaluating the ranking of the best result.
- Best result $\rightarrow$ most relevant document ranked at top position.
- Let r_i — rank at which the first relevant document of i retrieved.
 - ▶ Then reciprocal rank (RR) = $\frac{1}{r_i}$

$$\text{Mean reciprocal rank (MRR)} = \frac{\sum_{i=1}^n \frac{1}{r_i}}{n}$$

- Useful for *known-item* searches.
- $r_i \rightarrow$ rank at which the “answer” for query i retrieved.

Mean Reciprocal Rank

example

$$\text{Mean reciprocal rank (MRR)} = \frac{\sum_{i=1}^n \frac{1}{r_i}}{n}$$

- MRR=0.25

Mean Reciprocal Rank

example

$$\text{Mean reciprocal rank (MRR)} = \frac{\sum_{i=1}^n \frac{1}{r_i}}{n}$$

- MRR=0.25 $\rightarrow$ on average the system finds the known-item in rank 4.
- MRR= 0.75

Mean Reciprocal Rank

example

$$\text{Mean reciprocal rank (MRR)} = \frac{\sum_{i=1}^n \frac{1}{r_i}}{n}$$

- MRR=0.25 → on average the system finds the known-item in rank 4.
- MRR= 0.75 → on average the system finds the known-item between ranks 1 and 2.

Graded Relevance

- Binary Relevance:
 - ▶ Relevant 1
 - ▶ Non-relevant 0
- Some docs are *more* relevant to a query than other relevant ones!
 - ▶ We need non-binary relevance
- Graded relevance:
 - ▶ Perfect 4
 - ▶ Excellent 3
 - ▶ Good 2
 - ▶ Fair 1
 - ▶ Bad 0

Graded Relevance

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 - ▶ We need non-binary relevance
- Graded relevance:
 - ▶ Perfect 4
 - ▶ Excellent 3
 - ▶ Good 2
 - ▶ Fair 1
 - ▶ Bad 0
- Two assumptions:
 - ▶ Highly relevant documents are more useful than marginally relevant.
 - ▶ The lower the ranked position of a relevant document, the less useful it is for the user, since it is less likely to be examined.

Evaluation Metric with Graded Relevance

cumulated gain (Jarvelin and Kekalainen, ACM TOIS 20(4), 2002)

Idea:

- Highly relevant documents are more valuable than marginally relevant documents
- Documents ranked low are less valuable

Evaluation Metric with Graded Relevance

cumulated gain (Jarvelin and Kekalainen, ACM TOIS 20(4), 2002)

Idea:

- Highly relevant documents are more valuable than marginally relevant documents
- Documents ranked low are less valuable

$$Gain \in \{0, 1, 2, 3\}$$

$$G = \langle 3, 2, 3, 0, 0, 1, 2, 2, 3, 0, \dots \rangle$$

$$CG[i] = \sum_{j=1}^i G[j]$$

Normalized Discounted Cumulated Gain - (n)DCG

Jarvelin and Kekalainen, ACM TOIS 20(4), 2002

$$DCG[i] = \begin{cases} CG[i] & \text{if } i < b \\ DCG[i-1] + G[i]/\log_b i & \text{if } i \geq b \end{cases}$$

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$$DCG[i] = \begin{cases} CG[i] & \text{if } i < b \\ DCG[i-1] + G[i]/\log_b i & \text{if } i \geq b \end{cases}$$

$$\mathbf{Ideal} \ G = \langle 3, 3, 3, 2, 2, 1, 1, 0, \dots \rangle$$

$$nDCG[i] = \frac{DCG[i]}{\mathbf{Ideal} \ DCG[i]}$$

nDCG Calculation Example

Step-by-Step Walkthrough

Scenario

- Relevance grades: 0 (Bad), 1 (Fair), 2 (Good), 3 (Excellent)
- Actual ranking: [3, 2, 0, 3, 1] (relevance scores)
- Ideal ranking: [3, 3, 2, 1, 0] (sorted descending)

Actual DCG Calculation

$$DCG@1 = 3$$

$$DCG@2 = 3 + 2 / \log_2(2) = 5$$

$$DCG@3 = 5 + 0 / \log_2(3) = 5$$

$$DCG@4 = 5 + 3 / \log_2(4) = 6.5$$

$$DCG@5 = 6.5 + 1 / \log_2(5) = 6.93$$

Ideal DCG Calculation

$$iDCG@1 = 3$$

$$iDCG@2 = 3 + 3 / \log_2(2) = 6$$

$$iDCG@3 = 6 + 2 / \log_2(3) = 7.26$$

$$iDCG@4 = 7.26 + 1 / \log_2(4) = 7.76$$

$$iDCG@5 = 7.76 + 0 / \log_2(5) = 7.76$$

$$nDCG@5 = \frac{6.93}{7.76} = 0.893$$

Metric Selection Guide

Choosing the Right Tool for the Job

Use Case	MAP	nDCG	MRR	P@K
Academic Research	✓	✓		
Web Search		✓		✓
Known-Item Search			✓	
Legal E-Discovery	✓			
Recommendation		✓		
Early Prototyping				✓

Binary Relevance

- **MAP**: Comprehensive evaluation
- **P@K**: User experience focus
- **MRR**: Single answer tasks

Graded Relevance

- **nDCG**: Modern standard
- Captures result quality
- Web search favorite

Significance Testing

M1



M2



M3



The Problem

- System M1 has $\text{MAP} = 0.31$
- System M2 has $\text{MAP} = 0.29$
- ? Is M1 **really better** or is this just by chance?

Hypothesis Testing

- Check whether the difference can occur by chance?
- Null hypothesis (H_0): **M1** and **M2** are equivalent (no significant difference).
 - ▶ performance metrics of **M1** and **M2** random samples from same distribution.
- Compute probability of H_0 :
 - ▶ how probable that the values are from the same distribution?
 - ▶ If $P(H_0) \leq 5\% \rightarrow$ reject $H_0 \rightarrow$ **M1 significantly better than M2**.

Hypothesis Testing

Is the Difference Real?

Common Tests

- **t-test**: For normally distributed data
- **Wilcoxon signed-rank**: Non-parametric
- **Sign Test**: Simple but less powerful

Hypothesis Testing

Is the Difference Real?

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- **t-test**: For normally distributed data
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Best Practices

- Use **paired tests** (same queries)
- Report **p-values** and **confidence intervals**

Hypothesis Testing

Is the Difference Real?

Common Tests

- **t-test**: For normally distributed data
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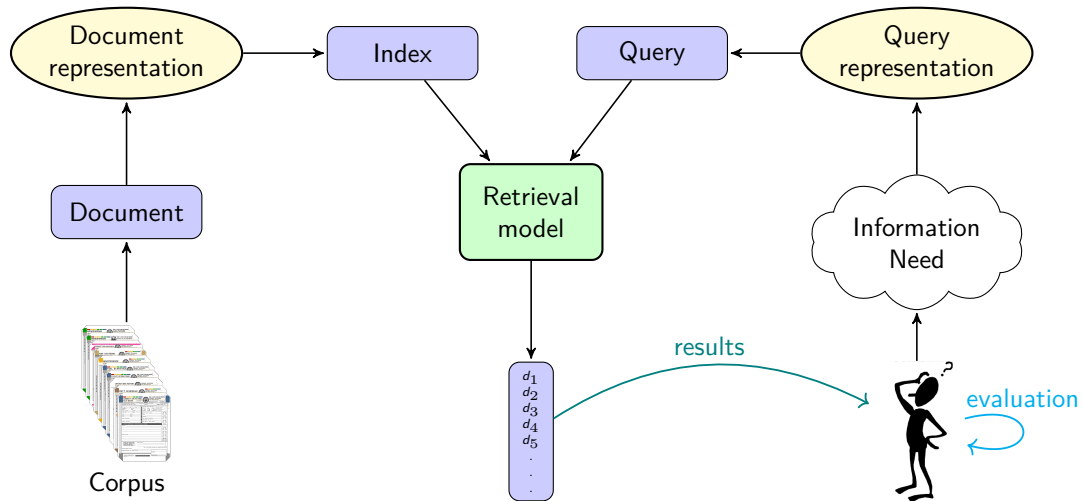
Best Practices

- Use **paired tests** (same queries)
- Report **p-values** and **confidence intervals**

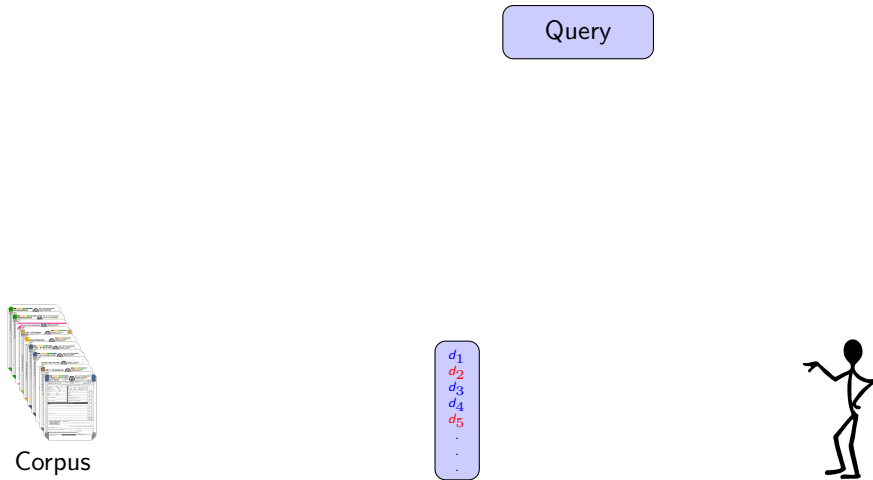
Interpretation

- **$p < 0.05$** : Statistically significant (95% confidence)
- **$p < 0.01$** : Highly significant (99% confidence)
- **$p > 0.05$** : Difference could be due to chance

Information Retrieval - A graphical representation



Information Retrieval - A graphical representation



Cranfield Method (CLEVERDON ET AL., 60s)

Benchmark data

- Document collection
- Query / topic collection
- Relevance judgments - information about which document is relevant

syllabus

question paper

correct answers

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TREC



<https://trec.nist.gov/>

CLEF



<https://www.clef-initiative.eu/>

NTCIR



<https://research.nii.ac.jp/ntcir>

FIRE



<https://fire.irsi.org.in/>

Cranfield Method (CLEVERDON ET AL., 60s)

Benchmark data

- Document collection
- Query / topic collection
- Relevance judgement

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question paper

They provide:

- Q - set of queries to **search with**;
- $\mathcal{D}$ - set of documents to **search into**;
- **Relevance judgement** of documents in $\mathcal{D}$ for each query of Q .

TREC



<https://trec.nist.gov/>



<https://www.clef-initiative.eu/>

NTCIR

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FIRE

INFORMATION RETRIEVAL
SOCIETY OF INDIA



<https://fire.irsii.org.in/>

Pooling

- Exhaustive relevance judgments may be infeasible.
- Pool top results obtained by various systems and assess the pool.
- *Unjudged documents are assumed to be non-relevant.*

Can unbiased, incomplete relevance judgments be used to reliably compare the relative effectiveness of different retrieval strategies?

- A wide variety of models, retrieval algorithms is important.
- **Manual interactive retrieval** is a must.

Assessor Agreement

Cohen's Kappa

- Judges / assessors may not agree about relevance.

Example:

	Yes ₁	No ₁	Total ₂
Yes ₂	300	20	320
No ₂	10	70	80
Total ₁	310	90	400

$$P(A) = (300 + 70)/400 = 370/400 = 0.925$$

$$P(\text{nrel}) = (80 + 90)/(400 + 400) = 0.2125$$

$$P(\text{rel}) = (320 + 310)/(400 + 400) = 0.7878$$

$$P(E) = P(\text{non-rel})^2 + P(\text{rel})^2 = 0.665$$

$$\kappa = \frac{P(A) - P(E)}{1 - P(E)} = \frac{0.925 - 0.665}{1 - 0.665} = 0.776$$

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$$\kappa = \frac{P(A) - P(E)}{1 - P(E)} = \frac{0.925 - 0.665}{1 - 0.665} = 0.776$$

- Rules of thumb:
 - $\kappa > 0.8$ — good agreement
 - $0.67 \leq \kappa \leq 0.8$ — fair agreement
 - $\kappa < 0.67$ — poor agreement

Experimenting with an actual Benchmark Dataset

FIRE

INFORMATION RETRIEVAL
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<https://fire.irsi.org.in/>

Data - <https://fire.irsi.org.in/fire/static/data>

Individual Access form - <https://fire.irsi.org.in/fire/download/organisational.pdf>

Individual Access form - <https://fire.irsi.org.in/fire/download/individual.pdf>

TREC Eval: The Standard IR Evaluation Tool

The Problem

- We have our ranked lists from different systems.
- We have the relevance judgments (qrels).
- Now we need to compute all those metrics we just learned.

The Solution: `trec_eval`¹

- Official evaluation software used in TREC since 1992.
- Command-line tool that computes 20+ different metrics.
- Open source and widely adopted in IR research.
- Handles large-scale evaluation automatically.

¹https://github.com/usnistgov/trec_eval, https://anaconda.org/conda-forge/trec_eval, https://huggingface.co/spaces/evaluate-metric/trec_eval

TREC Eval File Formats

Input Requirements

Run File (System Output)

Format:

```
query-id  Q0  doc-id  rank  score  RUN-ID
```

Example:

126	Q0	en.2.281.34.2009.9.11	1	14.693296432495117	BM25
126	Q0	en.3.326.205.2009.9.2	2	14.677349090576172	BM25
126	Q0	en.3.329.19.2009.9.22	3	14.648075103759766	BM25
126	Q0	en.13.25.179.2009.11.8	4	14.379230499267578	BM25

Qrels File (Ground Truth)

Format:

```
query-id 0 doc-id relevance
```

Example:

126	Q0	en.13.1.53.2009.5.31	0
126	Q0	en.13.10.164.2009.7.29	0
126	Q0	en.13.10.19.2009.7.30	1
126	Q0	en.13.25.179.2009.11.8	0

Relevance:

- 0 → Non-relevant
- 1 → Relevant

Running TREC Eval: Basic Usage

Command Line Syntax

```
trec_eval [options] qrels_file results_file
```

Basic Example

```
# Evaluate with all default metrics
```

```
trec_eval qrels.txt my_run.txt
```

```
# Evaluate specific metrics only
```

```
trec_eval -m map -m P.10 -m ndcg_cut.10 qrels.txt my_run.txt
```

Common Options

- `-m metric`: Specify which metric to compute
- `-q`: Output per-query results

TREC Eval Output: Understanding the Results

Sample Output

runid	LMDIR
num_queries	50
num_retrieved	12500
num_rel	450
num_rel_ret	380
map	0.3215
gm_map	0.2856
P_10	0.4560
ndcg	0.5234
ndcg_cut_10	0.6123
R-prec	0.4012
recip_rank	0.7834
bpref	0.3456

Key Sections

- **Run statistics:** Overview
- **Per-query results:** Detailed breakdown
- **Summary metrics:** Final scores

Important Fields

- **num_rel:** Total relevant docs
- **num_rel_ret:** Relevant docs retrieved
- **runid:** Your system identifier

Common TREC Eval Metrics

How to Specify Them

Basic Metrics

<code>-m map</code>	# Mean Average Precision
<code>-m gm_map</code>	# Geometric MAP
<code>-m P.10</code>	# Precision at 10
<code>-m P.20</code>	# Precision at 20
<code>-m ndcg</code>	# nDCG
<code>-m ndcg_cut.10</code>	# nDCG at 10
<code>-m Rprec</code>	# R-Precision
<code>-m recip_rank</code>	# Reciprocal Rank (MRR)

Recall-Oriented

<code>-m recall.1000</code>	# Recall at 1000
<code>-m iprec_at_rec.0.5</code>	# Interpolated P@recall=0.5

Advanced Metrics

<code>-m bpref</code>	# Binary preference
<code>-m infAP</code>	# Inferred AP
<code>-m instrec</code>	# INSTant RECall

Practical TREC Eval Example

Prepare Your Run File (6 column file)

```
# fire_lmdir_run.res
126    Q0      en.2.281.34.2009.9.11  1      17.589683532714844    LMDIR
126    Q0      en.3.329.19.2009.9.22  2      17.534711837768555    LMDIR
126    Q0      en.3.333.11.2009.10.24  3      17.205379486083984    LMDIR
126    Q0      en.3.326.205.2009.9.2   4      17.082340240478516    LMDIR
126    Q0      en.3.331.84.2009.10.10  5      16.845508575439453    LMDIR
...
```

Evaluation based on specific metrics

```
trec_eval -m map -m P.10 -m ndcg -m Rprec en.qrels.126-175.2011.txt fire_lmdir_run.res
```

Evaluation based on all metrics

```
trec_eval -m all_trec en.qrels.126-175.2011.txt fire_lmdir_run.res
```

Per-query performance analysis

```
trec_eval -m all_trec -q en.qrels.126-175.2011.txt fire_lmdir_run.res > results.txt
```


Interpreting TREC Eval Results

What Do the Numbers Mean?

Red Flags

- **MAP < 0.1**: System likely broken
- **P@10 = 0**: No relevant docs in top 10
- **num_rel_ret = 0**: Found no relevant docs
- **Huge variance**: Inconsistent performance

Comparative Analysis

System	MAP	P@10	nDCG
BM25	0.2987	0.4220	0.6278
LM-Dir	0.3129	0.4660	0.6486